INCH-POUND

MIL-T-81556A
AMENDMENT 1
26 September 1996

MILITARY SPECIFICATION

TITANIUM AND TITANIUM ALLOYS, EXTRUDED BARS AND SHAPES AIRCRAFT QUALITY

This amendment forms a part of MIL-T-81556A, dated 31 January 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

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2.1.1: Add the following, after MIL-STD-410:

"MIL-STD-2154 - INSPECTION, ULTRASONIC, WROUGHT METALS, PROCESS FOR"

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3.7: Add the following to the end of the sentence:

"and table Va."

3.10: Delete and substitute 3.10a and 3.10b as follows:

"a. The surface shall be free of contamination from oxygen (alpha case), hydrogen, nitrogen, or other surface contamination determined in accordance with 4.4.2.2. When permitted by the contracting agency, extrusions which will be machined all over may have an oxygen rich layer, 0.007 inch maximum, provided that such a layer is removable within the machining allowance.

b. The depth of surface imperfections (laps, die lines, dents, gouges, etc.) shall be contained within the dimensional tolerances and shall not extend beyond the minimum section thickness allowed per table VI."

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3.12: Delete and subsitute:

"3.12 Workmanship. The product, when visually examined (see 4.4.1), shall be uniform in quality and condition, sound, and free from cracks, seams, foreign materials and other defects detrimental to the fabrication or performance of parts. In addition, visual examination of transverse cross sections shall show no pipe or cracks; and shall show no segregation determined in accordance with 4.4.2.1. (See 3.10 for requirements regarding surface contamination.)"

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4.4.4.2: Delete and substitute:

"4.4.4.2 Inspection of billets. Ultrasonic inspection of billets may be performed in lieu of ultrasonic inspection of extruded product. The surface roughness shall not exceed 250 Ra. The billets shall conform to the ultrasonic quality requirements of 3.7."

4.4.4.3: Delete and substitute:

"4.4.4.3 Inspection of extruded product. Ultrasonic inspection of extruded product shall only be required if the billet has not been ultrasonically inspected. Ultrasonic inspection shall be applicable for extruded product 0.500 inch, or greater, in section thickness and a surface roughness no greater than 250 Ra. The extruded product shall conform to the ultrasonic quality requirements of 3.7."

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Table V: Delete and substitute with the following two tables:

Table V. Ultrasonic quality of extruded product.

Specified diameter - rod Specified thickness - bar Specified thickness - shape	Ultrasonic class 1/
0.500 - 1.500 inches 1.501 - 7.00 inches 7.01 inches and greater	AA A1 A

1/ Classification is in accordance with AMS 2631 (Grade 1 if double melted and Grade 2 if triple melted).

Table Va. Ultrasonic quality of billets.

Diameter or distance between flats, inches	Ultrasonic class	Specification
5.00 - 10.00 10.01 - 15.00 15.01 and above	A B C	AMS 2631 AMS 2631 MIL-STD-2154

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Table VI: In the first table at the top of the page, which contains the thickness tolerance, delete the fourth column entitled "Allowable depth of surface defects, inch".

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Table VIII: Delete and substitute:

Table VIII. Mechanical property specimen requirements (see 4.4.7.1).

Form	Thickness, diameter or distance between flats (inches)	Configuration 1/	Test location and direction 4/
Round, square, and hexagonal bar	0.187 - 1.500	D1, D2, D3 or D4 2/	Center of cross section
	1.501 and over	D1, D2, D3	Midway between the center and surface of the cross-section
Rectangular bar	0.187 - 0.375	F2 3/	Full Thickness 5/
	0.376 and over	D1, D2, or D3	Midway between the center and surface of the cross-section
Other Shapes	Unless otherwise specified (see 6.2.1), tension specimens shall be taken from the center of the cross section of the predominant area or thickest portion of product using the round specimens per note 2/ for product 1.500 inches and under in thickness. Full thickness rectangular specimens per note 3/ may be used for product 0.375 inch or less in thickness. Tensile specimens shall be taken midway between the center and surface for product over 1.500 inches in thickness.		

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Notes for table VIII.

1/ Configurations shown are for tension specimens. Compression specimens shall conform to the requirements of ASTM E 9.

2/ D1, D2, D3, and D4 designates specimens corresponding to round tension specimens as specified in ASTM E 8 for 0.500, 0.350, 0.250, and 0.160 inch nominal diameters in the reduced test section, respectively. In case of a dispute, the results obtained from testing the largest possible diameter of these four specimen sizes shall govern.

3/ F2 designates standard sheet type type specimens with a 1/2 inch width reduced test section as specifed in ASTM E 8 for rectangular tension specimens.

4/ Unless otherwise specified (see 6.2.1), the longitudinal axis of the tension specimens shall be parallel to the extruding direction and shall coincide with the specified test location within a radius of 1/16 inch.

5/ Machining the specimen thickness shall be permitted if it is limited to the removal of surface imperfections and surface contamination (see 3.10).

Custodians:
Army - MR
Navy - AS
Air Force - 11

Preparing Activity
Navy - AS

(Project 9540-0160)

Review Activities:
Army - AR, MI
Navy - OS, SH
Air Force - 99, 84
DLA - IS

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